



THE WALKER BUDDY

Grade 07

ALIGNED WITH INTERNATIONAL STANDARDS

This book supports essential educational frameworks that shape future-ready minds:

- 21st Century Skills
- Sustainable Development Goals (SDGs)
- Social-Emotional Learning (SEL)
- Universal Design for Learning (UDL)

STEAM Scouts is a ground-breaking STEAM book series with an interdisciplinary focus designed to ignite curiosity, creativity, and critical thinking in learners worldwide. The series supports Social-Emotional Learning (SEL) by promoting core competencies like self-awareness, self-management, social awareness, relationship skills, and responsible decision-making. It also aligns with the UN Sustainable Development Goals (SDGs), fosters 21st Century Skills, and meets international educational standards to equip learners with tools for a dynamic, innovative, and impact-driven future. Furthermore, it is built on Universal Design for Learning (UDL) principles, ensuring inclusive, accessible, and engaging STEAM education for all students, regardless of learning ability.

Key Features

Standards-Aligned

- NGSS (Next Generation Science Standards) – Authentic scientific inquiry & engineering design.
- CCSS (Common Core State Standards) – Rigorous literacy & math integration.
- Cambridge & IB (PYP/MYP/DP) – Global perspectives & interdisciplinary learning.
- ISTE (Technology Standards) – Digital literacy & computational thinking.
- STEL (Standards for Technological & Engineering Literacy) – Real-world problem-solving.

Social-Emotional Learning (SEL) and Global Citizenship

- Each book fosters collaboration, resilience, leadership, and ethical decision-making while connecting to multiple UN Sustainable Development Goals (SDGs) - such as Quality Education (SDG 4), Clean Energy (SDG 7), Climate Action (SDG 13), and Innovation (SDG 9) - empowering students to think globally, encourages ethical innovation & community impact.
- Differentiated challenges to meet diverse learning needs and skill levels.
- Visual cues and infographics to enhance understanding and engagement.
- Guided instruction to support learners through new or complex concepts.
- Scaffolded activities that build skills progressively and promote learner independence.

Why Choose This Series?

Future-Ready Skills – Prepares learners with essential skills like creativity, collaboration, and critical thinking for future careers and global challenges.

Teacher & Student-Friendly – Includes engaging lesson hooks, discussion prompts, guided instruction, and hands-on tasks for seamless learning.

- **SDG-Themed Modules** – Each module links to 2+ UN SDGs, blending academics with global purpose and ethical action.
- **Inclusive & Engaging** – Built on UDL principles to ensure accessibility through differentiated pathways, visual supports, and hands-on learning.

Perfect For:

- K-12 Schools adopting IB, Cambridge, NGSS, or CCSS
- STEM/STEAM Coordinators seeking industry-relevant curricula
- Homeschoolers and Educators wanting SDG-aligned, future-focused content
- Educational Institutions committed to sustainability and inclusion

Theme

03

THE WALKER BUDDY

Each week has a double session, total time of session 80-90 mins



Week	Unit Name	Lesson	Page
Week 1	Unit 1: Spark The Challenge!	Lesson 1: Feel the Walk.	2-20
		Lesson 2: EDP in Action.	21-31
Week 2		Lesson 3: Shape It Up in 3D!	32-51
Week 3	Unit 2: Innovators in Action.	Lesson 1: Feel the Force.	52-69
Week 4		Lesson 2: Less Effort, More Impact.	70-89
Week 5		Lesson3: Alert Mode On.	90-121
Week 6	Unit 3: Beyond Imagination.	Lesson 1: Print the Perfect Fit.	122-139
Week 7		Lesson 2: Tap to Help.	140-160
Week 8		Lesson 3: From Code to Care.	161-184

About The Theme

Every person deserves the freedom to move, explore, and live safely—regardless of age or ability. Yet as people grow older, simple daily tasks like walking, remembering routines, or staying connected can become more challenging. In this theme, students will take on the role of young engineers, coders, and designers, creating a Smart Walker that empowers older adults to remain mobile, independent, and safe through the power of technology.

Students will investigate how science, technology, and artificial intelligence can address real-world aging challenges. They'll design and build smart walker prototypes equipped with ultrasonic sensors, vibration motors, obstacle alerts, and custom-built components using tools like Tinkercad and Fusion 360. By working with materials such as PVC and real electronic components, they'll transform ideas into tangible, user-focused solutions.

With tools like Arduino and MIT App Inventor, students will explore the power of coding and connected devices. They'll write programs that trigger real-time safety alerts, remind users to take medications, and even send notifications to caregivers—turning the walker into a truly smart assistant. These features showcase how AI-powered reminders and context-aware alerts can enhance safety and autonomy for elderly users.

Throughout the project, students will engage in:

- Sensor integration and digital communication
- Human-centered design and empathy mapping
- Coding logic, conditionals, and event-based programming
- Introductory AI applications for health and safety

This theme connects directly with Sustainable Development Goals 3 (Good Health and Well-being) and 10 (Reduced Inequalities), encouraging students to think critically about innovation, care, and inclusion. By the end of the theme, students will have created a functional smart mobility aid—and developed a powerful understanding of how technology, empathy, and AI can come together to improve lives and build a more compassionate, tech-enabled future for all.

HOW TO USE THIS BOOK?

1

Ready, Set, Learn!

Where we merge your existing knowledge with exciting new discoveries.

2

Inquiry-Based Learning!

Let's Think!

In this section, students will brainstorm and explore possible solutions to specific problems. This encourages critical thinking and creativity as they consider various approaches.

Explorer's Toolkit

Here, students will dive deeper into the topic. They will engage with curated videos, articles, educational games, and hands-on experiments to broaden their understanding and spark curiosity.

DIY Lab

It's part of the explore, students will undertake hands-on projects where they will design, build, and test their own creations. They will apply scientific principles, use technology and engineering skills, explore artistic concepts, and utilize mathematical reasoning.

Screen Lessons

Students will watch videos that explain key concepts in an engaging and accessible way.

Dig Deeper

Students will read articles that provide deeper insights into specific concepts, enhancing their understanding.

3

Problem-Based Learning:

- **Introducing the problem:**
Students will join a journey to tackle real-world environmental challenges. Through hands-on activities and engaging tasks, they will apply their knowledge and skills to discover innovative solutions.
- **Apply the Engineering Design Process (EDP):**
Students will use a structured approach to define problems, conduct research, generate ideas, prototype solutions, test, evaluate, and refine their designs.
- **The Scientific Method:**
Students will follow a systematic process of inquiry—making observations, forming hypotheses, conducting experiments, analyzing data, and drawing conclusions—to develop a deeper understanding of scientific concepts and problem-solving. In doing so, they will incorporate the CER framework to present their scientific claims, provide evidence from their experiments, and explain their reasoning behind the conclusions they draw.

B

4

Project-Based Learning:

PBL immerses students in real-world challenges through hands-on projects. They build critical thinking, creativity, collaboration, and communication skills while designing solutions and taking ownership of their learning.

Case Study

The journey shifts from storytelling to real-world problem-solving. Through engaging case studies, students will analyze real-life challenges, explore innovative solutions, and apply STEAM principles to tackle complex issues. By investigating real-world scenarios, they will develop critical thinking, problem-solving, and decision-making skills, preparing them for future careers and global challenges.

5

Assessment Exercises:

These are divided into three categories to ensure comprehensive learning and mastery.

Apprentice:

Developing understanding and can apply skills with guidance.

Builder:

Demonstrates solid understanding and can apply skills independently.

Mastermind:

Demonstrates deep understanding and can apply skills creatively and strategically.

6

Career Readiness

This section equips students with essential skills for future careers by fostering problem-solving, teamwork, and adaptability. Through hands-on projects, real-world applications, and critical thinking challenges, students develop the confidence and competencies needed for success in the ever-evolving job market.

7

Mission Innovate

Now, it's your turn! In Mission Innovate; students will apply their creativity, problem-solving skills, and knowledge to develop their own competition projects. This challenge encourages independent thinking, collaboration, and real-world innovation, empowering students to become the next generation of creators and problem-solvers.

Now I Can

This section is designed to help you measure your progress and achievements. Enjoy your exploration and learning journey through this STEAM book!

My Journal

In this part students will write their own story while they are creating the project.

By the end of this theme, students will be able to:

- Identify and describe common daily challenges experienced by elderly people.
- Demonstrate empathy by observing and reflecting on the needs of elderly individuals.
- Research and evaluate existing tools and technologies designed to assist elderly people.
- Work through the Engineering Design Process to create and improve solutions.
- Understand what mechanical engineers and robotics engineers do and the roles they play.
- Explore Fusion 360 and learn how to create basic 3D designs using tools like Sketch, Extrude, and Modify to develop functional parts smart walker.
- Explain how Newton's Third Law applies to walking and walker design.
- Calculate work, power, and efficiency using formulas:
$$\text{Work} = \text{Force} \times \text{Distance},$$
$$\text{Power} = \text{Work} \div \text{Time},$$
$$\text{Efficiency} = \text{Output} \div \text{Input} \times 100.$$
- Analyse prototype data using tables to identify design improvements and efficiency trends.
- Evaluate and choose materials based on strength, weight, and grip.
- Understand how body measurements and ergonomics affect design.
- Use AI tools to accurately scale real-life body measurements for a prototype, demonstrating an understanding of how scaling factors are applied in design and modeling.
- Adapt walker designs for different environments and user needs.
- Use Arduino to connect and control motion or obstacle detection sensors.
- Build a basic electronic circuit that detects obstacles and sends vibration or sound alerts.
- Design an emergency alarm housing, then export, slice, and prepare their 3D model using Cura for printing as part of their Smart Walker prototype.
- Design a user-friendly emergency alert app by understanding how components work, organizing UI elements clearly, and applying basic UX principles.
- Learn how to use the Clock component in MIT App Inventor.
- Create a reminder app that helps elderly users remember to drink water and take their medicine.

Unit

1

Lesson

1

Spark The Challenge! Feel the Walk.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Identify and describe common daily challenges experienced by elderly people.
- Demonstrate empathy by observing and reflecting on the needs of elderly individuals
- Research and evaluate existing tools and technologies designed to assist elderly people
- Create an empathy map to visualize what elderly users say, do, think, and feel, using insights from interviews, observations, or stories to define user needs and guide design decisions.

Challenge questions of the lesson

- What small everyday tasks might become difficult as people get older, and why do you think that happens?
- How might life feel different from the perspective of an elderly person, and what can we do to better understand their experience?
- What smart tools or inventions already exist to support older adults, and how well do they actually solve their problems?
- If you stepped into the shoes of an elderly person, what would you say, do, think, and feel during a regular day—and how can this help you design something useful for them?

SEL Relation

- **Social Awareness:** Students will enhance their ability to empathize with others by understanding the experiences of elderly individuals.
- **Relationship Skills:** Engaging in discussions and collaborative activities fosters communication and cooperation among peers.

Engage



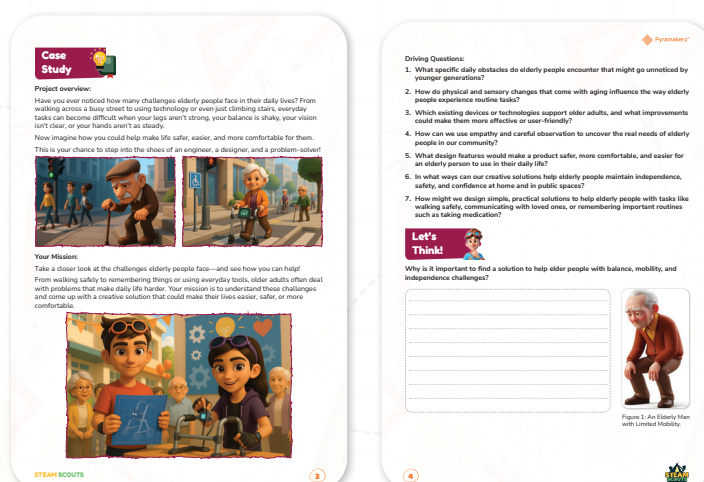
Teacher's Role:

- Introduce the scenario through an engaging discussion that connects to students' experiences and prior knowledge.
- Incorporate interactive strategies such as thought-provoking visuals, storytelling elements, and open-ended questions to ignite curiosity and foster engagement.
- Encourage students to make predictions by prompting them to analyze clues from the title, images, or key concepts, helping to develop critical thinking skills.
- Facilitate meaningful discussions by creating a supportive environment where students feel confident to share their ideas, ask questions, and consider diverse perspectives.
- Provide targeted support by guiding discussions with scaffolding techniques, ensuring that all students can participate and contribute their unique viewpoints.

Expected From Students:

- Actively contribute to discussions by sharing thoughts, asking questions, and making thoughtful predictions based on the story.
- Engage with multimedia elements such as visuals, sounds, and thought-provoking prompts to deepen understanding and curiosity.
- Express personal ideas confidently while listening respectfully to peers, fostering a collaborative and inclusive learning environment.
- Make meaningful connections by reflecting on the scenario themes and relating them to personal experiences and real-world situations.
- Demonstrate critical thinking by analyzing scenario elements, considering different perspectives, and drawing insightful conclusions.

Parts of the book included.



Case Study



1. Read the Case Study Together

- Display and read the Project Overview aloud with expression.
- Pause at the line “This is your chance to step into the shoes of an engineer, a designer, and a problem-solver!”
- Ask: “What does it mean to ‘step into someone else’s shoes?’” to introduce empathy.

2. Introduce the Mission

- Read Your Mission.
- Ask students to brainstorm one problem an elderly person might face in their daily life and share with the class.
- Note down ideas on the board.

3. Guide Inquiry with Driving Questions

- Display the Driving Questions list.
- Choose 1–2 questions to discuss as a whole group. For example:
 - “Which existing devices help older adults, and how could we improve them?”
- Encourage students to think beyond mobility — include communication, safety, and comfort.

4. Set the Challenge

- Explain that over the coming lessons, they will:
 - Identify a specific problem.
 - Use empathy mapping and observation.
 - Brainstorm, design, and prototype a creative solution.
- Emphasize: “We’re not just learning about technology — we’re using it to improve real lives.”

Possible Strategies for Implementation:

1. Hook & Empathy-Building Activity

- Strategy: Begin with an emotional and experiential hook to foster empathy.
- Implementation:
 - Show a short video or real-life story of an elderly person facing mobility or memory challenges.
 - Lead a classroom discussion: “What was difficult for the person? How did it make you feel?”
 - Have students complete a “walk in their shoes” simulation: e.g., wear gloves to mimic reduced dexterity, or carry weights to simulate weak muscles.

2. Perspective-Taking Activity

- Strategy: Movement Simulation
- Implementation: Have students simulate aging-related mobility issues by walking slowly, wearing gloves, or carrying something heavy. Then ask:
 - “How did that feel?” - “What everyday tasks became harder?”

3. Storytelling as a Tool

- Strategy: Frame the project as a story where students are the heroes solving a real problem.
- Implementation:
 - Introduce the scenario using a storytelling format: “You’re part of a young engineering team designing a smart walker for someone’s grandma.”
 - Give students a design journal titled “My Engineering Journey” where they’ll document their thoughts, empathy notes, and ideas.

4. Role-Playing Interviews

- Strategy: Engage students in role-playing to understand user needs.
- Implementation:
 - Pair students to conduct mock interviews—one plays an elderly person, the other a student engineer.
 - Provide prompt cards with common elderly challenges to help guide the interaction.
 - Reflect together: “What did you learn about the needs and feelings of your user?”

5. Gallery Walk: Aging Challenges

- Strategy: Visual and collaborative activity to categorize and discuss problems.
- Implementation:
 - Place posters or stations around the room (e.g., “Mobility,” “Memory,” “Vision,” “Technology”).
 - Students walk around and post sticky notes with challenges or examples they’ve seen.
 - As a class, categorize the input and vote on which area they want to focus their project on.

6. Driving Question Launch Pad

- Strategy: Use the driving questions to spark curiosity and define the challenge.
- Implementation:
 - Display the key questions (e.g., “What specific daily obstacles do elderly people encounter that might go unnoticed by younger generations?”).
 - Create a question wall or digital whiteboard where students post ideas and curiosities.
 - Break students into small groups to brainstorm observations or experiences related to elderly people (family members, neighbors, etc.).

Possible answers for driving questions:

(They will just discuss it in the class).

1. **What specific daily obstacles do elderly people encounter that might go unnoticed by younger generations?**
 - Mobility difficulties such as walking, climbing stairs, and balance issues
 - Sensory impairments like poor eyesight and hearing loss
 - Memory challenges and cognitive decline (e.g., dementia)
 - Difficulty with daily tasks like cooking, dressing, and medication management
 - Feelings of loneliness and social isolation
 - Managing chronic health conditions like arthritis, diabetes, or heart disease
2. **How do physical and sensory changes that come with aging influence the way elderly people experience routine tasks?**
 - Weaker muscles and joint stiffness make movement slower and more painful
 - Reduced vision and hearing affect communication and safety
 - Cognitive decline can cause forgetfulness and difficulty following instructions
 - Balance issues increase the risk of falls and injuries
 - These changes can make simple tasks like walking, eating, or using technology more challenging
3. **Which existing devices or technologies support older adults, and what improvements could make them more effective or user-friendly?**
 - Mobility aids such as canes, walkers, and wheelchairs
 - Hearing aids and large-print or magnifying devices
 - Reminder systems for medication and appointments
 - Emergency alert systems and wearable health monitors
 - Improvements could include easier usability, better comfort, integration with smart home systems, and personalized features to match individual needs
4. **How can we use empathy and careful observation to uncover the real needs of elderly people in our community?**
 - By listening to their experiences and concerns without assumptions
 - Observing their daily routines to identify hidden challenges
 - Engaging in conversations that explore emotional as well as physical needs
 - Using empathy maps to understand what they say, think, feel, and do
 - Involving elderly people directly in the design and testing of solutions
5. **What design features would make a product safer, more comfortable, and easier for an elderly person to use in their daily life?**
 - Ergonomic handles and grips to reduce strain

- Lightweight but sturdy materials for ease of use and durability
- Clear, large, and simple controls or displays
- Stability features to prevent falls or tipping
- Adjustable components to fit different body sizes and abilities
- Non-slip surfaces and easy-to-clean materials

6. In what ways can our creative solutions help elderly people maintain independence, safety, and confidence at home and in public spaces?

- Enabling safer mobility through smart walkers or canes with sensors
- Providing reminders and alerts to support memory and medication adherence
- Facilitating communication with caregivers and family via easy-to-use devices
- Enhancing home safety with automated lighting or fall detection systems
- Encouraging social interaction through accessible technology and community programs

7. How might we design simple, practical solutions to help elderly people with tasks like walking safely, communicating with loved ones, or remembering important routines such as taking medication?

- Incorporate sensors and alarms in walking aids to detect obstacles or falls
- Develop voice-activated or large-button communication devices
- Create wearable reminders that gently alert users to take medications
- Design intuitive apps or devices with minimal buttons and clear instructions
- Use modular designs that allow customization based on individual needs

7. Mission Launch & Problem Framing

- Strategy: Frame the project as a real-world design mission.
- Implementation:
 - Formally launch the mission with a “design brief” handout summarizing the challenge.
 - Guide students to write their own mission statement: “I want to help elderly people by designing a solution that...”
 - Use a team charter activity: teams agree on roles, collaboration norms, and goals.

Let's Think!



Objective:

Students will think critically about the challenges elderly people face in maintaining balance, mobility, and independence. They will reflect on the importance of finding solutions to these challenges and how their designs can make a positive impact on improving the lives of older adults.

Possible Answers:

- Because it helps them stay safe, move around more easily.
- Maintain their dignity and quality of life.
- Support their independence
- Reduces the risk of falls
- Allows them to stay active and connected to others.

Possible Strategies for implementation:

1. Strategy: Socratic Questioning

- Implementation:
 - Lead a class discussion using probing questions like:
 1. “What could happen if an older person loses their ability to move around safely?”
 2. “How does mobility impact someone’s sense of freedom or dignity?”
 - Encourage students to support their answers with examples or evidence from real life, or observations.

2. Strategy: Reflective Journaling

- Implementation:
 - Ask students to write a short reflection on the prompt:
“Why is it important to help elderly people stay mobile and independent?”
 - Encourage them to make personal connections or imagine their own grandparents in this situation.
 - Share responses in pairs or groups for deeper discussion.

3. Strategy: Role Reversal Discussion

- Implementation:
 - Have students imagine they are 80 years old and facing daily mobility struggles.
 - Ask: “How would it feel to rely on others for everything?”
 - Discuss how balance and independence are tied to dignity, mental health, and quality of life.

4. Strategy: Visual Brainstorm or Mind Mapping

- Implementation:
 - In small groups, students create a mind map around the question:
“Why should we care about mobility and independence in older adults?”
 - Branch out to related ideas like: safety, mental health, social inclusion, healthcare costs.
 - Groups can share their maps with the class.

5. Strategy: Real-World Connection

- Implementation:
 - Show a short news clip or article about mobility issues in the elderly.
 - Discuss the impact on their lives and on their families or communities.
 - Ask: “What does this show us about why finding a solution is important?”

6. Strategy: Scenario Analysis

- Implementation:
 - Present students with different real-life or fictional scenarios (e.g., a senior who can’t climb stairs, or who falls often).
 - Have them discuss in small groups:
 1. “What are the risks in this situation?”
 2. “What changes could improve their life?”
 - This helps students analyze the problem critically and practically.

Explorer’s Toolkit



Teacher’s Role:

- **Questioner:** Uses prompts like:
 - “What did you notice about how it felt to complete that task?”
 - “What else could make this easier for someone with limited movement or vision?”
- **Supporter:** Models reflective thinking and encourages the use of sentence starters:
 - “It was hard when...”
 - “I felt frustrated because...”

Provides scaffolding for students who need help expressing feelings or design ideas.
- **Collaborator:** Circulates during the design phase, encouraging teamwork and combining ideas. Promotes inclusive collaboration, especially for students who identified different empathy insights.
- **Assessor:** Observes teamwork, listens for thoughtful reflections, and asks:
 - “How does your design solve one of the challenges we experienced?”
 - “Who would benefit most from your solution?”

- **Technology Integrator (Optional):**

- Supports students in documenting their prototypes through photos or videos and showcasing their process digitally on class boards or reflection journals.

Expected From Students:

- Participate respectfully and follow instructions
- Try each challenge thoughtfully (not for fun, but to understand)
- Reflect honestly on how they felt doing the tasks
- Share their observations and emotions in discussion or writing

Parts of the book included.



DIY Lab

Objective:

Students will gain empathy by experiencing activities that simulate the challenges elderly people face, helping them understand difficulties with mobility, dexterity, and vision. This will inform their design solutions for the Smart Walker.

Materials Needed:

- Open space for pairs to interact
- Optional: Picture cards or emotion flashcards for support

Activity Instructions:

1. Wear a pair of thick gloves and try to zip your jacket or open a pencil case.
2. Put on sunglasses indoors and try to read a short message or label.
3. Slowly sit down and stand up as if your legs feel heavy or sore.

Possible Strategies for Implementation:

- **Teamwork:** Have students pair up to try these activities together and discuss their findings afterward.
- **Reflection:** Guide students to ask themselves how they felt during each activity and how it might relate to the design of a walker for elderly people.
- **Empathy Mapping:** Use the activities to create an empathy map for elderly users.

Possible Answers:

- What part was tricky or uncomfortable?
 - “Zipping the jacket was hard because the gloves limited my finger movement.”
 - “Reading with sunglasses was difficult because I couldn’t see clearly.”
 - “Getting up from the chair was tough because my legs felt heavy.”
- How would I feel if I had to do this many times every day?
 - “I would feel frustrated and tired if I had to do this every day. It would make simple tasks harder and take longer.”
 - “It would be exhausting to keep trying when it’s so difficult to see or move quickly.”

Explain



Teacher’s Role:

- Facilitate a discussion about personal experiences and perceptions of aging and accessibility.
- Introduce and guide the video activity by explaining its purpose and relevance, encouraging students to observe emotions, physical struggles, and environmental obstacles.
- Encourage teamwork and monitor student discussions

Expected From Students:

- Watch real-life videos of older adults facing mobility issues.
- Participate in-group discussion.
- (Optional) Write a short reflection on what surprised them the most about the mobility challenges older adults face.

Parts of the book included.



Screen Lessons

To understand more about mobility challenges and what makes walking difficult for some older adults, let's watch the following video till the 2:55.

<https://www.webmd.com/healthy-aging/video/older-adults-mobility>



Screen Lessons

Objective:

Students will watch a video to understand the mobility challenges older adults face and how those challenges impact walking. They will then apply this knowledge to design solutions that improve safety and mobility for elderly users.

Video link: <https://www.webmd.com/healthy-aging/video/older-adults-mobility>



Possible Strategies for implementation:

1. Pre-Watching Strategies

Strategy: Set a Purpose for Watching

- How to implement:
 - Explain the goal: “We’re watching this video to understand real-life mobility challenges older adults face.”
 - Give a focus question: “What makes walking difficult for some older adults?”

Strategy: Quick-Write (Empathy Warm-Up)

- How to implement:
 - Ask students to write for 2–3 minutes on: “Have you ever seen or helped an older person who had difficulty walking? What did you notice or feel?”
- Optional: share in pairs to build emotional connection before watching.

Strategy: Vocabulary Preview (Optional)

- How to implement:
 - Introduce key terms students may hear in the video, like: Mobility, joint stiffness, balance issues, independence
 - Write definitions on the board or in a handout.

2. During Watching Strategies

Strategy: Guided Note-Taking

- How to implement:
 - Provide students with a note-taking worksheet that includes:
 1. Mobility Challenge Observed
 2. Impact on Daily Life
 3. Emotional Response
 - Pause at key moments to allow students to jot down notes.

Strategy: Focused Viewing with Empathy Lens

- How to implement:
 - Remind students to watch from the perspective of the person in the video: “What might they be feeling in this moment?”
 - Let them track one individual’s journey and jot down:
 1. What they say
 2. What they do
 3. What they struggle with

Strategy: Pause and Reflect

- How to implement:
 - Pause the video at powerful moments and ask:
 1. “What did you just notice?”
 2. “How might that feel if it were you or someone in your family?”

3. Post-Watching Strategies

Strategy: Think-Pair-Share

- How to implement:
 - Ask students to respond to:
 1. “What stood out to you the most in the video?”
 2. “How did it make you feel?”
 3. “What challenges surprised you?”
 - Think silently → discuss in pairs → share with the class.

Strategy: Reflection Prompt

- How to implement:
 - Give students 5 minutes to respond in writing to:
“Why is it important to find a solution for mobility challenges in older adults?”
 - Optional: Collect reflections or use them to start a follow-up discussion.

Strategy: Real-Life Connection

- How to implement:
 - Ask: “Have you seen any of these challenges in your own family or community?”

Encourage students to share examples, drawing connections between the video, the scenario, and the world around them.

Elaborate and Evaluate



Teacher's Role:

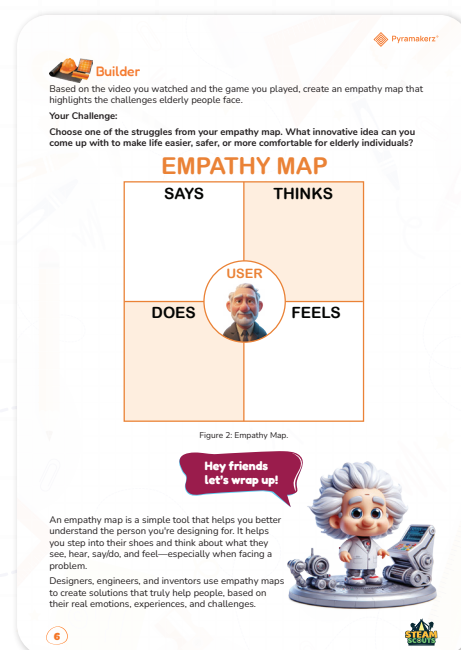
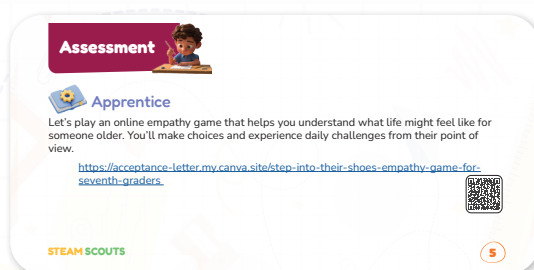
- **Supporter:** Models how to record insights using sentence starters like:
 - “It was difficult to...”
 - “This made me feel...”Supports diverse learners by allowing sketching, verbal reflection, or physical demonstration.

- **Collaborator:**
 - Encourages shared reflections after each part of the lab and game. Joins group discussions to model empathy and guide idea-building.
- **Assessor:**
 - Observes student behaviors and conversations for signs of empathy, thoughtful design, and STEM reasoning. Uses observation checklists and questioning during building and feedback rounds.
- **Facilitator:**
 - Leads the empathy map activity, guiding students to reflect on what elderly individuals might feel, say, do, and think after watching the video.
- **Collaborator:**
 - Encourages teamwork by promoting inclusive group discussions and ensuring that all voices are heard.
- **Observer:**
 - Monitors group interactions to assess engagement, understanding, and the depth of student reflections.

Expected From Students:

- Participate in physical and digital empathy simulations.
- Share reflections using thinking routines.
- Identify daily challenges of elderly people.
- Collaboratively build a simple machine to solve a selected challenge.
- Test, revise, and justify design choices.
- Reflect on user experience and suggest improvements.

Parts of the book included.





Apprentice

Objective:

Understand the emotional and physical challenges elderly people experience in daily tasks. Use simulation and gameplay to build compassion and design insight.

Resource: <https://acceptance-letter.my.canva.site/step-into-their-shoes-empathy-game-for-seventh-graders>



Possible Strategies for Implementation:

1. Strategy: Classroom Setup

- **Arrange Seating for Interaction:** Organize students in small groups or circles to encourage discussion and participation.
- **Establish Ground Rules:** Emphasize respect, active listening, and confidentiality to create a safe environment for sharing and role-play.

2. Game Introduction

- **Contextualize Empathy:** Begin with a brief discussion or story about empathy—what it means and why it matters in daily life and school settings.

3. Gameplay Strategies

- **Role-Playing:** Assign or let students choose characters from the game. Encourage them to act out scenarios, focusing on understanding and expressing their character's feelings and perspectives.

4. Facilitate Discussion:

- After each scenario, guide students in discussing how their character felt, what challenges they faced, and how they might respond empathetically.
- **Use Inquiry-Based Prompts:** Ask open-ended questions such as, "How would you feel in this situation?" or "What could you do to help someone in this scenario?" to deepen reflection.



Builder

Objective:

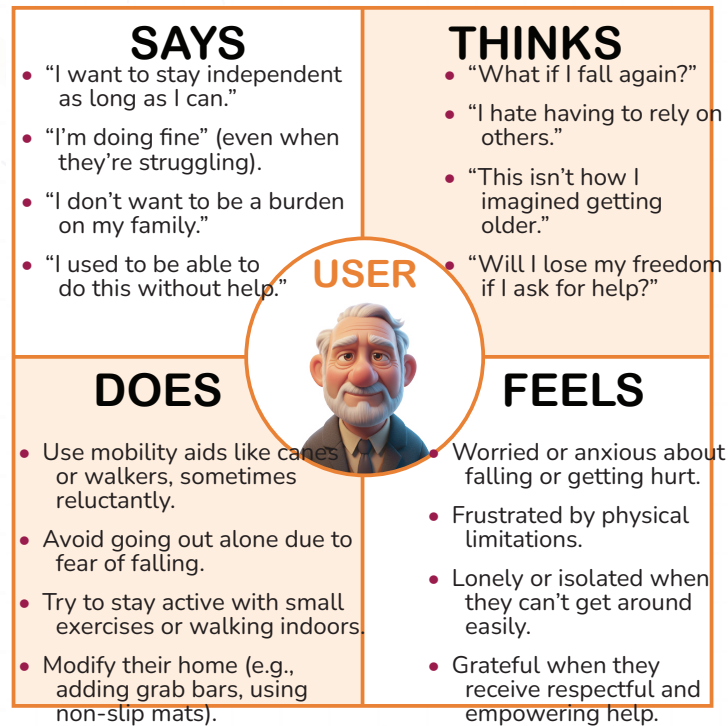
- Analyze challenges elderly individuals face by creating an empathy map based on a video and interactive game.
- Generate an innovative idea or solution that addresses one of those challenges, focusing on safety, comfort, or accessibility.

Extra Resource: <https://www.nngroup.com/articles/empathy-mapping/>



Possible Answers:

EMPATHY MAP



Materials Needed:

- Printed or digital empathy map template (with sections: Says, Thinks, Does, Feels)
- Video (previously watched)
- Game reflection sheet (optional)
- Writing tools (pens, pencils, markers)
- Sticky notes or small slips of paper (optional for brainstorming)
- Large poster paper or whiteboard for group sharing (optional)

Activity Instructions:

1. Recall & Reflect:

- Think back to the video and empathy game. What physical, emotional, or cognitive challenges did elderly individuals face?

2. Empathy Mapping:

- Fill in each section of the empathy map:
 - SAYS – What did the person say or might say aloud?
 - THINKS – What are they likely thinking but not saying?
 - DOES – What actions or behaviors did you observe?
 - FEELS – What emotions were visible or likely felt?

3. Challenge Prompt:

- Review the struggles captured in your empathy map.

- Choose one major challenge (e.g., bending down, reading small text, feeling isolated).

4. Innovate & Solve:

- Design or brainstorm one innovative idea or product that could make life easier, safer, or more comfortable for the elderly individual based on that challenge.
- Sketch or describe your idea in 2–3 sentences.

5. Share:

- Present your idea to a partner, group, or class.
- Explain how your idea is informed by what you learned through the empathy process.

Possible Strategies for Implementation:

1. Empathy Mapping Session:

- **Group Work:** Have students work in pairs or small groups to discuss and fill out the empathy map. Encourage them to think about the elderly person's daily life and challenges they might face.
- **Guided Questions:** Provide guiding questions to help students fill in the empathy map, such as "What would an elderly person say when they have trouble walking?" or "What are they thinking when they struggle to complete daily tasks?"

2. Design Brainstorming:

- **Idea Sketching:** After identifying a challenge from the empathy map, have students sketch their innovative design solutions to make life easier for elderly individuals.
- **Mind Mapping:** Students can use a mind map to expand on their ideas and think about potential features for their design, such as how it would help with mobility, safety, or comfort.

3. Prototype Creation:

- **Low-Fidelity Prototypes:** Allow students to build simple prototypes using materials like paper, cardboard, or digital tools to visualize their solutions. This will help them express their ideas more clearly.

4. Class Discussion and Reflection:

- **Group Presentations:** Have students present their ideas to the class, explaining how their design addresses a specific challenge and how it enhances the life of an elderly individual.
- **Class Reflection:** Encourage the class to discuss each solution and consider how it could be improved or made more accessible.

5. Real-World Examples:

- **Research Existing Solutions:** Ask students to research existing technologies or designs that help elderly people (such as walkers, hearing aids, or medication reminders) and use those ideas to inspire their own designs.



Now I Can

Objective:

Students will reflect on their learning journey throughout the lesson, identifying and articulating their mastery of key concepts related to empathy, inclusive observation, and user-centered design. This includes recognizing the invisible challenges people face, practicing interview skills, and using empathy maps to drive innovation.

Activity Instructions:

Step 1: Revisit the Lesson

Students review their work from the lesson, including empathy maps, observations, interview notes, and design sketches.

They consider:

- What did I learn about the challenges elderly people face?
- How did observing, listening, or role-playing help me understand someone else's perspective?
- What design idea did I create, and what inspired it?

Step 2: Student Self-Reflection

Students individually reflect on their learning using sentence starters such as:

- “One thing I now understand about empathy is...”
 - “A challenge I didn’t notice before this activity was...”
 - “My design idea helps because...”
- They may write, draw, or use a traffic light self-assessment:
I understand this well | I need more practice | I’m still unsure

Step 3: Small Group Discussion

In groups of 3–4, students share one insight from their reflection.

Use a Round Robin format to ensure each voice is heard.

Discussion prompts may include:

- “What surprised you during today’s activity?”
- “Which idea from your empathy map would help someone the most?”
- “How can we use empathy in other STEAM projects?”

Relation to Standards

1. Next Generation Science Standards (NGSS) – Science & Engineering Practices

- **SEP 1 – Asking Questions and Defining Problems:** Students identify real-world problems elderly people face with mobility.
- **SEP 3 – Planning and Carrying Out Investigations:** Through observation, simulation, and video analysis, students investigate aging and accessibility.

- **SEP 7 – Engaging in Argument from Evidence:** Students use evidence from videos/articles to support their empathy maps and design ideas.

2. ISTE Standards for Students (Technology & Digital Learning)

- **1. Empowered Learner:** Students set personal goals for understanding human-centered challenges and propose solutions.
- **4. Innovative Designer:** Empathy mapping is part of a design process that starts with defining user needs.
- **6. Creative Communicator:** Students organize their observations and feelings into visual empathy maps and class discussions.

3. Common Core ELA Standards – Speaking & Listening / Informational Text

- **CCSS.ELA-LITERACY.SL.7.1:** Students engage in collaborative conversations, share insights, and respond to peer ideas.
- **CCSS.ELA-LITERACY.RI.7.1 & RI.7.7:** Students read an article and watch a video to gather relevant evidence about aging and mobility.
- **CCSS.ELA-LITERACY.W.7.9:** Optional reflection or journal prompt links writing to informational sources.

4. C3 Framework (Social Studies/Empathy Focus)

- **D2.Civ.10.6-8** – Explain the origins, functions, and structure of systems that support people's needs.
This applies when students consider how society supports aging populations and how design can play a role.
- **D4.6.6-8** – Draw on multiple disciplinary lenses to analyze problems.
Students look at aging from physical, emotional, and social perspectives.

5. 21st Century Learning & Career Readiness (P21 Framework)

- **Critical Thinking & Problem Solving:** Students explore the problem of limited mobility and how it impacts quality of life.
- **Collaboration:** Students work in pairs/groups to complete empathy maps and discuss user challenges.